

Meta Flowers

An Analogy of Life in the XR Era

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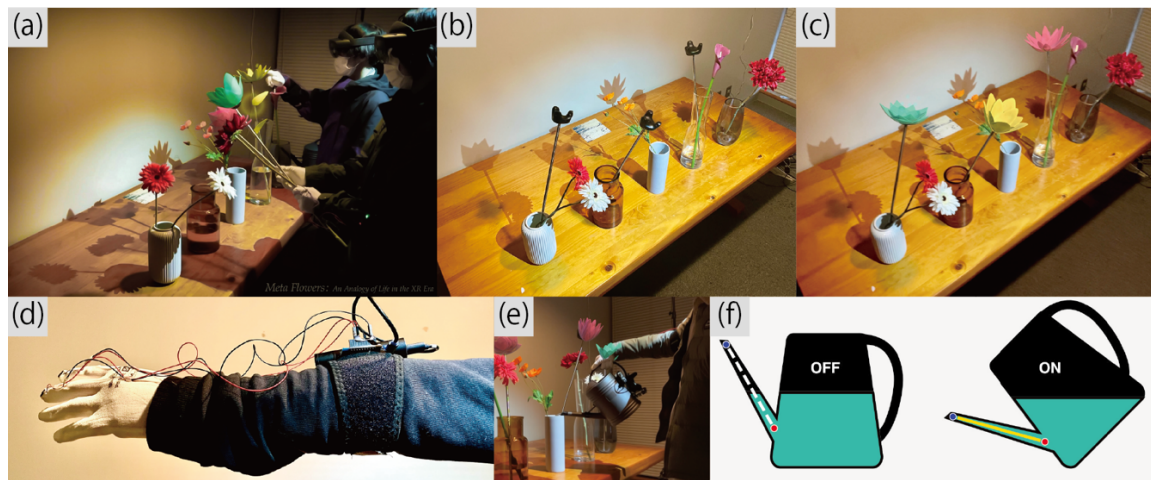


Figure 1: (a) Overview of the experience. (b) The Installation status as seen by naked eye. (c) The Installation status as seen through HoloLens 2. (d) The glove-type tactile device. (e) Water is poured into a vase with a metal rod without petals, using a sensor-equipped jug, the VF blooms. (f) When the jug is tilted to pour water, the electrodes are energized by the water and detect this condition.

ABSTRACT

“Meta Flowers” is a multiparticipant installation artwork with Cross Reality (XR). Wearing a glove-type tactile device consisting of linear resonant actuators (LRA) and a microcontroller with Wi-Fi unit, and HoloLens 2, participants can experience XR through the act of arranging virtual flowers (VFs) which have ‘shadow,’ ‘rigid-body,’ and ‘sound (see Figure 1 (a)).’ The VF blooms at the location of the VIVE Tracker, which is installed at the tip of a metal rod inserted into a vase on the table (see Figure 1 (b) (c)). Real shadows of the vases, sticks, and artificial flowers, as well as the virtual shadows

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of the VFs, are projected onto the table by the light and images projected from the projector installed on the ceiling, creating a fusion that is not unnatural. Participants can move the VFs and arrange them in vases on the table. When participants touch the VF, petals of the VF fall and participants get tactile sensations through the haptics glove (see Figure 1 (d)). VFs play sounds while they are in bloom, and their pitch change depending on the position of them. In the other words, the sound and appearance of VFs will change depending on the relationship among participants and VFs. In addition, when real water is poured by the sensor-equipped jug into a vase in which a metal rod without petals is put, the VF blooms again (see Figure 1 (e) (f)).

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts.

KEYWORDS

Cross Reality, XR, MR, Interactive Art, Haptics, HoloLens 2, VIVE Tracker

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1 INTRODUCTION

We have been working on the DSMR (Drawing Sound in MR Space) project, which aims to develop audiovisual art experiences in MR space. “Encounters,” one of the two most recent works in this project, focuses on the collision of physical and virtual [Nakagawa and Sonobe, 2019]. Participants can make a collision sound by hitting a physical object with a fired virtual bullet. At the same time, a virtual visual effect of a color determined by the pitch of the collision sound is generated. This enables an XR experience where the physical and virtual worlds intersect: hand gesture: physical → bullet: virtual → collision: physical × virtual → collision sound: physical → visual effect: virtual. “Encounters 2.0,” the sequel to Encounters, is a work that adds an experience focused on “light” to the Encounters experience [Nakagawa and Sonobe, 2020]. By colliding a virtual bullet with a light bulb (Light unit), the visual effect of the bulb breaking and the sound of it breaking are played simultaneously, and the light of the bulb disappears. In addition, a virtual sphere of light (ghost object) emerges from the Light unit turned off. The ghost object can be grasped by gestures, and by bringing it close to the Light unit in the extinguished state, the ghost object is sucked into the Light unit, and light returns to the light bulb. There are three bulbs, each with a different color of light. It is also possible to change the color of the light of the bulbs by passing through the ghost objects.

2 MOTIVATION

While working on these projects, we came to think that "light" is very important in the XR experience. In Encounters 2.0, we implemented light as something that moves seamlessly between the virtual and physical worlds, but the ghost object did not have the properties of a light source. Even though participants did not feel much discomfort and seemed to perceive the ghost object as light. Therefore, we thought that by giving the virtual light the characteristics of a light source, we could realize an even higher level of XR.

3 IMPLEMENTATION

"VR Shadow" by Naemura et al. uses a projector equipped with a 3D sensor as a light source to project a shadow of a virtual object in physical space [Naemura et al., 2002]. In "RV Shadow" by the same authors, shadows are generated by a 3D-sensor-equipped pseudo-light source and physical objects, and projected on a wall using a projector. In the artwork "Invisible Sculpture," which Tsubokura presented in 2017 [Tsubokura, 2017], invisible objects are illuminated by a pseudo-light source and shadows are generated with

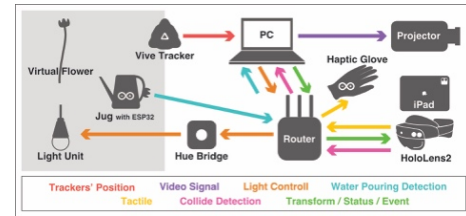


Figure 2: Communication among equipment.

a projector using a method that combines the above VR Shadow and RV Shadow. Referring to these examples, we created the Meta Flowers by mixing virtual flowers and real artificial flowers and constructing a method to merge their shadows (Figure 2).

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